(KTXFI1EBNF) Physics I. Practice

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May 4, 2026

Current Updates and Announcements

- May 4, and go. . .
- Homework No. 3 is now available tonight. The deadline is May 10, 2026 at midnight.
- I promised a sample exam. I provided $\sim$100 questions on Moodle.
- Today there is only practice of three different type of exercises.

Exercises I

- **Thermodynamics I:** Temperature scales, thermal expansion, thermometers. State variables, ideal gas, state changes, ideal gas equation of state, Boyle's law, Gay-Lussac's laws, universal (combined) gas law, heat, specific heat, molar heat, heat capacity.
- **Thermodynamics II:** Expansion work, internal energy, the first law of thermodynamics, forms of the first law in different state changes.
- **Thermodynamics III:** Cyclic processes, Carnot cycle, Carnot's theorem, reduced heat, entropy, the second law of thermodynamics.
- **Thermodynamics IV:** Kinetic theory of gases, molecular thermodynamics. Distributions, distribution functions. Elementary statistics.
- **Electrostatics I.:** Electric charge: Fundamental electrical phenomena, electric charge and matter, Coulomb's law (force between two point charges, unit of electric charge, permittivity of vacuum).

Exercises II

- **Electrostatics II.:** The electric field: The electrostatic field (electric field strength, electric field lines). Determination of electric field strength. Gauss's law (flux of the electric field). Electric field of a point charge.
- **Electrostatics III.:** Electrostatic potential: Work of the electric field (work done in an electric field). Electric potential (electrostatic potential). Electric potential difference, voltage. Electric potential of charge systems at small and large distances. Electrostatic energy.
- **Elements of physical or wave optics.** Branches of optics. Light interference, light diffraction, diffraction through a slit, optical grating, light polarization. Light as an electromagnetic wave: Poynting vector. Optical instruments.

Oscillations and Waves I

1. Given the displacement-time function of the oscillation:

$$y(t) = 1.6\sin(1.3t - 0.75)\ cm.$$

At $t = 0\,s$, determine:

a) displacement,
b) velocity,
c) acceleration.

Convert amplitude: $1.6\ \mathrm{cm} = 0.016\ \mathrm{m}$

a) $y(0) = 0.016\sin(-0.75) = -0.0109\ \mathrm{m}$
b) $v(t) = 0.016 \cdot 1.3\cos(1.3t - 0.75)\ v(0) = 0.0208\cos(-0.75) = 0.0152\ \mathrm{m/s}$
c) $a(t) = -0.016 \cdot (1.3)^2\sin(1.3t - 0.75)\ a(0) = 0.0184\ \mathrm{m/s^2}$

Oscillations and Waves II

2. Given the sum of two one-directional oscillation as the following $x(t) = 3\sin(\omega t) + 4\cos(\omega t)$. This describes the displacement - time function of the oscillation, where t represents time, ω denotes the circular frequency of the oscillation, and ω is a constant.

   a) Prove that the above x(t) oscillation, given by the displacement-time function, performs simple harmonic motion.

   b) What is the value of the resultant oscillation given by the above displacement-time function?

   a) Write in form $x(t) = A\sin(\omega t + \varphi)$
   $A = \sqrt{3^2 + 4^2} = 5$ (same unit as x)
   Thus it is simple harmonic motion.

   b) $x(t) = 5\sin(\omega t + \varphi)$, $\tan\varphi = \frac{4}{3}$

Oscillations and Waves III

3. Consider three one-directional oscillations with equal frequencies, whose amplitudes and initial phases are given as follows:

$$\begin{aligned} A_1 &= 7 \cdot 10^{-3}, \mathrm{m} \quad , \quad \varphi_1 = \pi/3 \\ A_2 &= 5 \cdot 10^{-3}\,\mathrm{m} \quad , \quad \varphi_2 = 5\pi/6 \\ A_3 &= 2 \cdot 10^{-3}\,\mathrm{m} \quad , \quad \varphi_3 = 23\pi/18 \end{aligned}$$

Let's combine these three oscillations. Calculate the amplitude and initial phase of the resultant vibration!

$X = \sum A_i \cos \varphi_i \approx -2.36 \cdot 10^{-3}\ \mathrm{m}$
$Y = \sum A_i \sin \varphi_i \approx 7.27 \cdot 10^{-3}\ \mathrm{m}$
$A = \sqrt{X^2 + Y^2} \approx 7.65 \cdot 10^{-3}\ \mathrm{m}$
$\varphi \approx 1.89\ \mathrm{rad}$

Oscillations and Waves IV

4. Calculate and determine the curve described by the resultant of the perpendicular vibrations given by the equations:

$$
\begin{aligned}
x(t) &= x_0 \cos(\omega t), \\
y(t) &= y_0 \cos(2\omega t).
\end{aligned}
$$

$\cos(2\omega t) = 2\cos^2(\omega t) - 1$
$y = y_0 \left(2(x/x_0)^2 - 1\right)$
Parabolic curve (dimensionless relation between x/x_0 and y/y_0).

Oscillations and Waves V

5. Let $f(\omega t \pm kx + \varphi)$ be a function whose argument is at least twice differentiable, continuous, and arbitrary, where ω: constant circular frequency, k : wavenumber (constant), ϕ initial phase (constant), t : time, x spatial coordinate. Prove that the function f described above is a solution to the one-dimensional wave equation.

Let $u = \omega t \pm kx + \varphi$

$\frac{\partial^2 f}{\partial t^2} = \omega^2 f''(u)$

$\frac{\partial^2 f}{\partial x^2} = k^2 f''(u)$

Thus:

$\frac{\partial^2 f}{\partial t^2} = \frac{\omega^2}{k^2} \frac{\partial^2 f}{\partial x^2}$

Wave speed: $v = \frac{\omega}{k}\ \mathrm{m/s}$

Oscillations and Waves VI

6. The displacement of a wave as a function of time is given by

$$y(x, t) = 0.27\sin(12x - 500t)\,mm,$$

where x is in meters and t is given in seconds. Determine at the point x = 0.2 meters in the case of a transverse wave:

a) its velocity,
b) its acceleration!
c) What is the displacement of the point at t = 4s seconds?

Convert amplitude: $0.27\ \mathrm{mm} = 2.7 \cdot 10^{-4}\ \mathrm{m}$

a) $v = -2.7 \cdot 10^{-4} \cdot 500\cos(12x - 500t)$ $v = -0.135\cos(2.4 - 500t)\ \mathrm{m/s}$
b) $a = -2.7 \cdot 10^{-4} \cdot 500^2\sin(12x - 500t)$ $a = -67.5\sin(2.4 - 500t)\ \mathrm{m/s^2}$
c) $y(4) \approx 2.6 \cdot 10^{-4}\ \mathrm{m}$

Rigid Bodies I

1. According to measurements, the position (in meters) of a body with a mass of 300 g along the x-axis is given by the following function:

$$x(t) = -t^2 + 7.5t^3,$$

where t represents the time in seconds. Determine the resultant force acting on the body as a function of time! What is the magnitude of the resultant force at t = 5 s?

Rigid Bodies II

$$
\begin{aligned}
v(t) &= \frac{dx}{dt} = -2t + 22.5t^2 \\
a(t) &= \frac{d^2x}{dt^2} = -2 + 45t \\
F(t) &= ma(t) = 0.3(-2 + 45t) \\
F(t) &= -0.6 + 13.5t \ \mathrm{N} \\
F(5) &= -0.6 + 67.5 = 66.9 \ \mathrm{N}
\end{aligned}
$$

Rigid Bodies III

2. A mass of 20 kg moves with a constant velocity at 12 m/s . At some moment, we start to apply a force opposite to the direction of motion, which is not constant but increases steadily from zero at a rate of 40 N/s . How much time elapses from the beginning of the force's effect until the body comes to a complete stop (decelerates)? A 20 kg mass moves with velocity 12 $\mathrm{m/s}$.

$$
\begin{aligned}
F(t) &= 40t \\
a(t) &= -\frac{F}{m} = -2t \\
v(t) &= 12 - \int 2t\,dt = 12 - t^2
\end{aligned}
$$

Rigid Bodies IV

Stop when $v = 0\, m/s$:

$$t^2 = 12 \Rightarrow t = \sqrt{12} = 3.46\,\mathrm{s}$$

Rigid Bodies V

3. A point-like object is subjected to a force in the x direction:

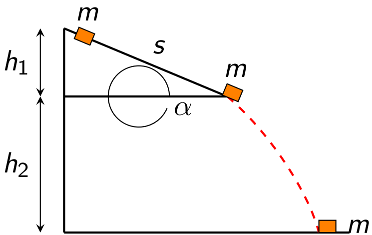
$$F(x) = (3 + 0.5x)\,\mathrm{N}.$$

Determine the work done by the force as the mass point moves from x=0 m to x=4 m.

$$\begin{aligned} W &= \int_0^4 (3 + 0.5x)\,dx \\ W &= \left[3x + 0.25x^2\right]_0^4 \\ W &= 12 + 4 = 16\ \mathrm{J} \end{aligned}$$

Rigid Bodies VI

4. A 0.2 kg mass stone slides without initial velocity from the top edge of a frictionless roof as depicted in the figure. $\alpha = 40^\circ$, g=10 m$/s^2$, h1=3 m, h2=15 m
   a) What will be the magnitude of the velocity at the moment of impact with the ground?
   b) What will be the velocity upon impact with the ground if there is friction on the slope and the coefficient of kinetic friction is 0.4.



Rigid Bodies VII

a) Energy conservation:

$$v = \sqrt{2gh_2} = \sqrt{300} = 17.32 \text{ m/s}$$

b) Work of friction:

$$W_f = \mu mg \cos\alpha \cdot \frac{h_1}{\sin\alpha}$$

Effective energy:

$$\begin{aligned} v &= \sqrt{2g\left(h_2 - \mu h_1 \cot\alpha\right)} \\ v &\approx 15.8 \text{ m/s} \end{aligned}$$

Rigid Bodies VIII

5. From a water reservoir, water flows to a turbine located 100.5 m below. 2.8 m^3 of water passes through the turbine per minute. The efficiency of the turbine is 80%. Neglecting air resistance and the minimal kinetic energy of the water leaving the turbine, calculate the power output of the turbine! The density of water is 1000 kg$/m^3$, g=9.8 m$/s^2$.

$$\begin{aligned}
\dot{V} &= \frac{2.8}{60} = 0.0467\ \mathrm{m}^3/\mathrm{s} \\
\dot{m} &= 1000 \cdot 0.0467 = 46.7\ \mathrm{kg/s} \\
P &= \eta \dot{m} g h \\
P &= 0.8 \cdot 46.7\, kg/s \cdot 9.8\, m/s^2 \cdot 100.5\, m \\
P &\approx 3.68 \cdot 10^4\ \mathrm{W}
\end{aligned}$$

Rigid Bodies IX

6. A vehicle traveling with velocity v on a horizontal road decelerates uniformly over a distance s. What slope angle α will keep the vehicle at rest in its braked state on a similarly graded road? Neglecting air resistance.

   From kinematics:

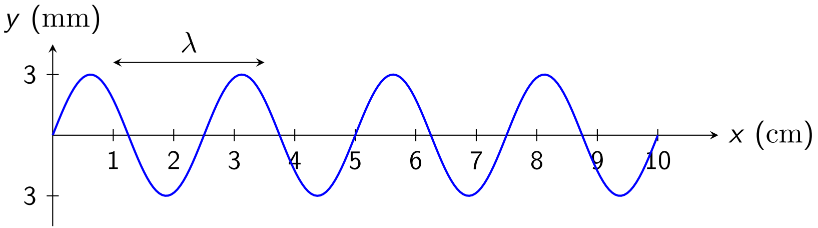
$$a = \frac{v^2}{2s}$$

   For equilibrium on slope:

$$\begin{aligned} g \sin\alpha &= a \\ \sin\alpha &= \frac{v^2}{2gs} \\ \alpha &= \arcsin\left(\frac{v^2}{2gs}\right) \end{aligned}$$

Wave Optics I

1. The wave shown in the figure was generated by a 60 Hz resonator. Determine:
   a) amplitude of the wave
   b) frequency of the wave
   c) wavelength of the wave
   d) propagation speed of the wave
   e) period of the wave



Wave Optics II

$$\begin{aligned}
A &= 3 \cdot 10^{-3}\,\mathrm{m} \\
f &= 60\,\mathrm{Hz} \\
\lambda &= 2.5\,\mathrm{cm} = 0.025\,\mathrm{m} \\
v &= f\lambda = 1.5\,\mathrm{m/s} \\
T &= \frac{1}{f} = 0.0167\ \mathrm{s}
\end{aligned}$$

Wave Optics III

2. A wave propagating along a string can be defined as follows (with x in meters, t in seconds): $y = 0.02\,m \sin(30t - 4x)$. Determine the:
   a) amplitude
   b) frequency
   c) speed
   d) wavelength

$$
\begin{aligned}
A &= 0.02\,\mathrm{m} \\
\omega &= 30\,\mathrm{rad/s} \\
k &= 4\,\mathrm{rad/m} \\
f &= \frac{\omega}{2\pi} = 4.77\,\mathrm{Hz} \\
v &= \frac{\omega}{k} = 7.5\,\mathrm{m/s} \\
\lambda &= \frac{2\pi}{k} = 1.57\,\mathrm{m}
\end{aligned}
$$

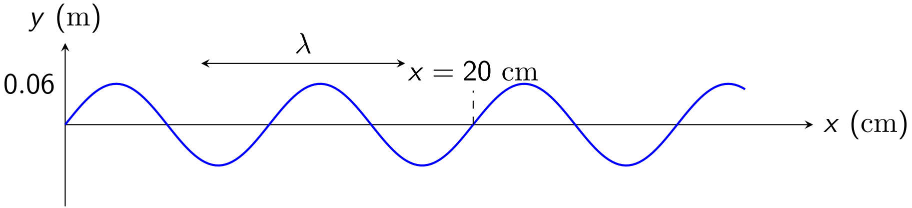
Wave Optics IV

3. Consider a wave given by the equation $f(x,t) = Ae^{-B(x-vt)^2}$. Prove that this wave equation satisfies the general one-dimensional wave equation.

   Substitute $u = x - vt$, then both second derivatives give the same structure, hence wave equation satisfied.

Wave Optics V

4. Determine the amplitude, frequency, and wavelength of the wave shown in the figure below, if the propagation speed of the wave is 300 m/s. Write a solution for the wave if at t=0 s the wave appears as shown in the figure.



Wave Optics VI

$$
\begin{aligned}
A &= 0.06\,\mathrm{m} \\
\lambda &= 0.03\,\mathrm{m} \\
f &= \frac{v}{\lambda} = 10000\,\mathrm{Hz} \\
\omega &= 2\pi f \\
k &= \frac{2\pi}{\lambda}
\end{aligned}
$$

Wave:

$$
y(x,t) = 0.06\sin(kx - \omega t)
$$

Wave Optics VII

5. For a wave propagating along a string with a frequency of 30 Hz, $\lambda = 60\,\mathrm{cm}$, A $=$ 2 mm, write the solution in SI units.

$$\begin{aligned} \omega &= 60\pi\,\mathrm{rad/s} \\ k &= \frac{2\pi}{0.6} \\ y(x,t) &= 2\cdot 10^{-3}\sin(kx - \omega t) \end{aligned}$$

Wave Optics VIII

6. A wave solution is given by $y = 0.27 \sin(12x - 500t)\, mm$, where x is in meters and t is in seconds. For the transverse wave at x=20 cm, determine:
   a) its speed
   b) its acceleration
   c) What is the displacement of the point at t=4 s?

$$\begin{aligned}
A &= 2.7 \cdot 10^{-4}\,\mathrm{m} \\
v &= \frac{500}{12} = 41.67\,\mathrm{m/s} \\
v(t) &= -0.135 \cos(2.4 - 500t)\,\mathrm{m/s} \\
a(t) &= -67.5 \sin(2.4 - 500t)\,\mathrm{m/s^2} \\
y(4) &\approx 2.6 \cdot 10^{-4}\,\mathrm{m}
\end{aligned}$$

The End

Thank you for your attention!